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Studierichting: Informatica
Oefeningenreeks 4A nr. 49.8

$$\int \frac{\sin(2x)}{1+\sin^2(x)}dx$$
$$= \int \frac{\sin(2x)}{1+\frac{1-\cos(2x)}{2}}dx$$
$$= 2\int \frac{\sin(2x)}{3-\cos(2x)}dx$$
$$u = 3-\cos(2x) \Rightarrow du = 2\sin(2x)dx$$
$$= \int \frac{du}{u}$$
$$= \ln|3-\cos(2x)| + c$$

OF

$$\int \frac{\sin(2x)}{1+\sin^2(x)}dx$$
$$= \int \frac{2\sin(x)\cos(x)}{1+\sin^2(x)}dx$$
$$u = \sin(x) \Rightarrow du = \cos(x)dx$$
$$= 2\int \frac{udu}{1+u^2}$$
$$v = 1+u^2 \Rightarrow dv = 2udu$$
$$= \int \frac{dv}{v}$$
$$= \ln|1+\sin^2(x)| + c$$

References